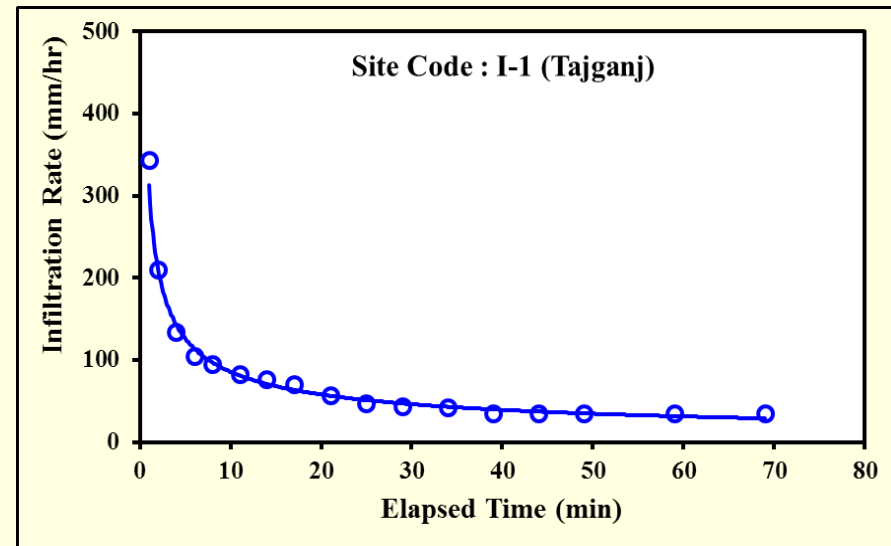
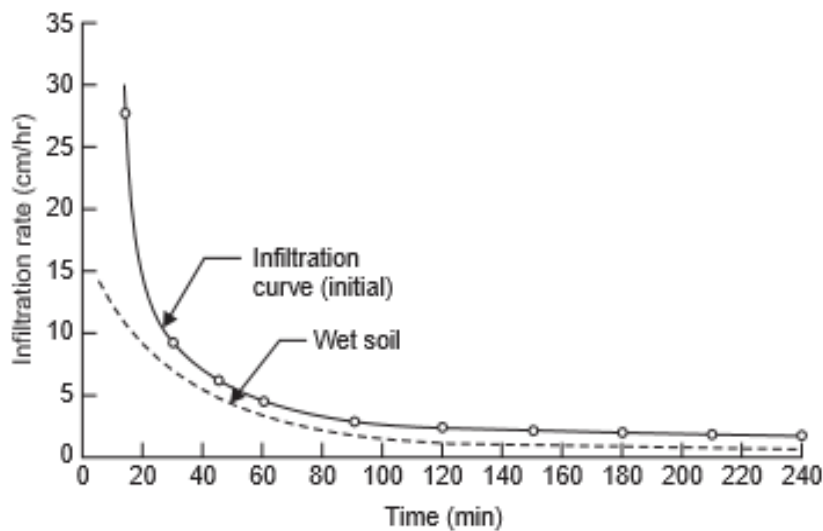


Infiltration

Infiltration is the process by which water on the ground surface enters into the soil. The velocity or speed at which water enters into the soil is called the infiltration rate. Infiltration is the measure of the ability with which a soil is able to absorb water. Technically, infiltration rate is the volume flux of water flowing into the soil profile per unit of soil surface area at given time.

In dry soil, water infiltrates rapidly. This is called the initial infiltration rate. The maximum rate at which the soil in any given condition is capable of absorbing water is called its **infiltration capacity (f_p)**.

As more water replaces the air in the pores, the water from the soil surface infiltrates more slowly and eventually reaches a steady rate. The infiltration rate decreases to a steady or quasi-steady state as the soil becomes saturated. Therefore, the filtration rate curve shows a sharp decline with time. The constant rate approached after a sufficiently large time is referred as the steady-infiltration rate.





Factors Control Infiltration

Soil Characteristics

Condition of soil surface

Vegetation Cover

Current moisture content

Soil temperature

Importance of Infiltration:

Infiltration rate is an important component of the hydrologic cycle. It helps to determine how rainfall is divided between recharging the groundwater and running off over the surface as sheetwash and in streams.

EVAPORATION

Evaporation is nothing but a thermal separation process, Theoretically evaporation means simply Vaporisation from the surface of a liquid, Vaporisation of liquid below its boiling point is called evaporation. Thus no boiling occurs and the rate of vaporization depends on the diffusion of vapour through the boundary layers above the liquid



Evaporation

- Evaporation happens in several ways

- Open water evaporation
- Transpiration from leaves
- Evaporation from soil and land surface

Evapotranspiration

evapotranspiration summarizes all processes that return liquid water back into water vapor

- **evaporation (E)**: direct transfer of water from open water bodies or soil surfaces
- **transpiration (T)**: indirect transfer of water from root-stomatal system
 - of the water taken up by plants, ~95% is returned to the atmosphere through their stomata (only 5% is turned into biomass!)

Factors Affecting the evaporation

1. Temperature
2. Surface area
3. Humidity
4. Wind speed

As the temperature of air rises, its capacity to hold moisture increases. An increase in temperature raises the temperature of water, which means more energy is available to the water molecule for escaping from the liquid to a gaseous state.

The rate of evaporation is proportional to the difference between the Saturation vapor pressure at the water temperature and the actual vapour pressure in the air.

Wind helps in removing the evaporated water from the zone of evaporation. Rate of evaporation increases with wind speed up to a critical speed. The critical wind speed value is a function of the size of the water surface.

STREAMFLOW MEASUREMENT AND HYDROGRAPH ANALYSIS

Stream flow is one of the most important topics in engineering hydrology because it directly relate to water supply, flood control, reservoir design, navigation, irrigation, drainage, water quality, and others.

Streamflow or discharge is a volume of water per unit of time. The standard units for measurement of discharge are m^3/s (cubic metres per second or cumecs) If we rewrite the units of discharge we can think of them as a water velocity (m/s) passing through a cross-sectional area (m^2). Therefore:

$$\text{m}^3/\text{s} = \text{m}/\text{s} \times \text{m}^2$$

How do we measure how much water in a stream?

- Volumetric measurements-
 - Work on very low flows, collect a known volume of water for a known period of time
Volume/time is discharge or Q
- Cross-section/velocity measurements
- Dilution gaging with salt or dye
- Artificial controls like weirs

A *hydrograph* is a graph showing discharge (i.e., stream flow at the concentration point) versus time. The various components of a natural hydrograph are shown in Fig. 5.1. At the beginning, there is only base flow (i.e., the ground water contribution to the stream) gradually depleting in an exponential form. After the storm commences, the initial losses like interception and infiltration are met and then the surface flow begins. The hydrograph gradually rises

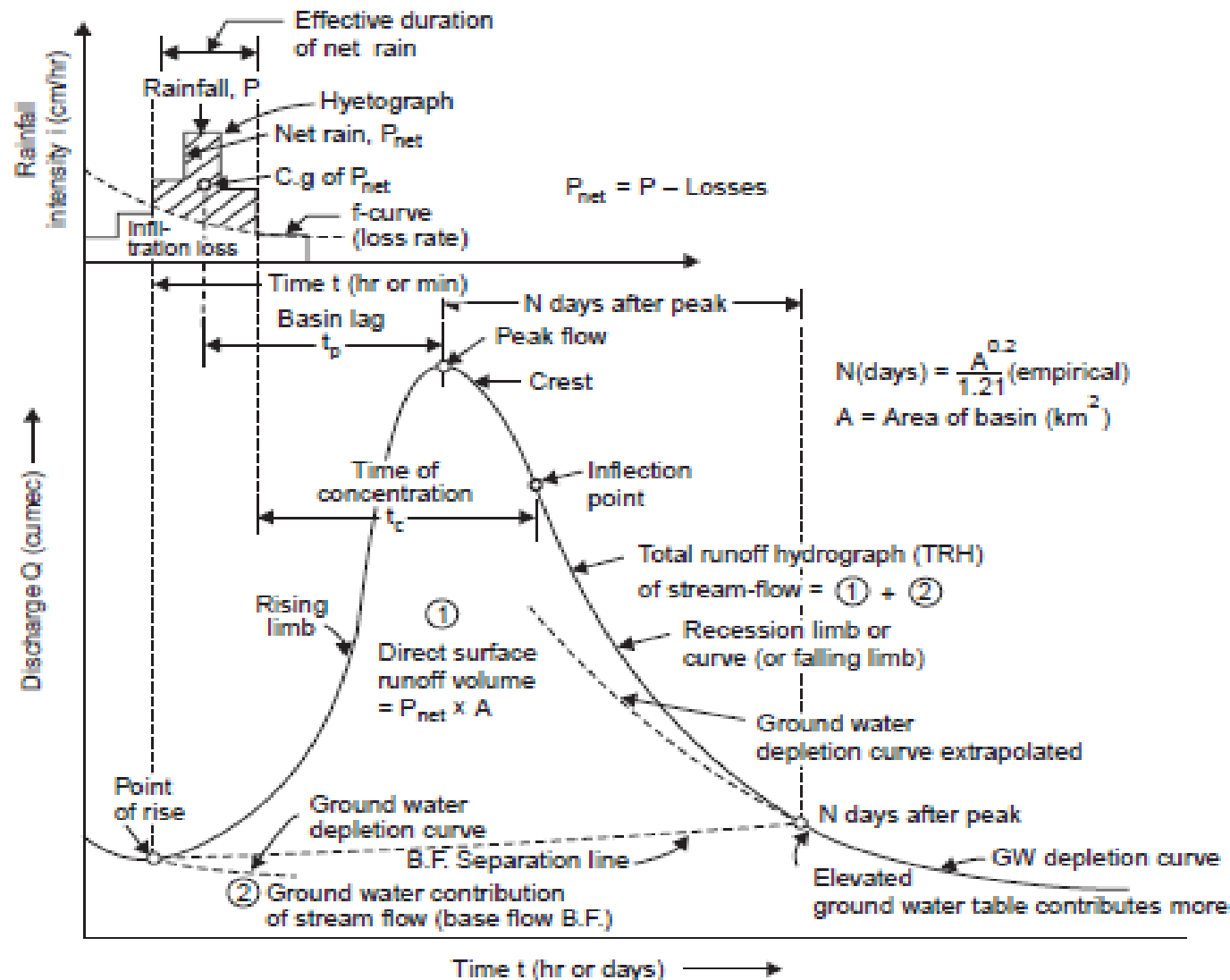
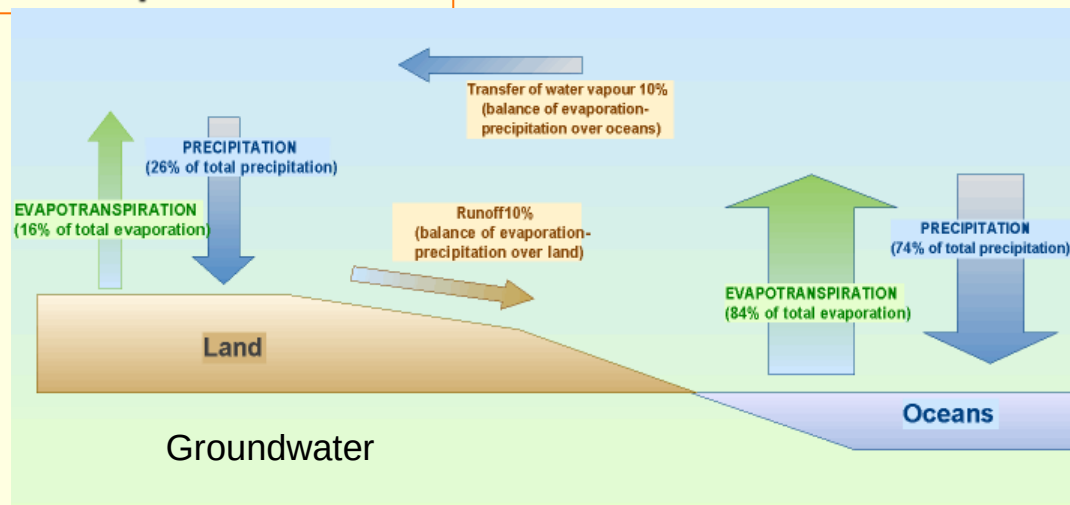


Fig. 5.1 Components of streamflow hydrograph

Processes moving water through the cycle... key Atmosphere Transport

What percent of the water in the atmosphere comes from evaporation?

- Evaporation 90%
- Transpiration 10%



SAVE WATER

